

Recognize

A breakthrough meeting intelligence system for Google Meet that identifies multiple speakers from a single camera and microphone — no voice enrollment or special hardware required.

- Identifies who is speaking in real time from standard meeting audio and video
- Captures and transcribes every conversation without setup or training
- Builds a persistent memory of past meetings from ordinary devices



What Recognize Solves

The Diarization Problem — Solved with a Single Camera and Mic

Recognize's breakthrough innovation is real-time multi-speaker attribution from just **one camera and one microphone**. No microphone arrays. No voice profiles. No enrollment. No dedicated hardware. That's what makes it different from classical diarization approaches, which have long relied on arrays, pre-recorded voice models, or high-latency batch cloud services. Recognize correlates **per-face lip movement (MAR variance)** with Whisper word timestamps, attributing speech without ever analyzing voice signatures.

The Institutional Memory Problem — Solved Structurally

Organizations lose knowledge every time a meeting ends. Recognize treats the **knowledge graph as the durable artifact** — not the transcript. Every meeting and document automatically joins the same persistent relational structure. Entities deduplicate. Relationships accumulate weight. Communities of related concepts emerge bottom-up, forming a living map of **what the organization actually thinks about**.

Part 1: Real-Time Speaker Diarization

A Chrome extension captures the Meet tab's audio and video via `chrome.tabCapture`, streaming raw PCM and JPEG frames over WebSocket to a local Python server — no external hardware required.

MediaPipe detects face landmarks and tracks lip movement to estimate when each speaker is talking. Whisper transcribes the audio with word-level timestamps, and the system matches those words to the lip-motion timeline to attribute speech to the right speaker.

- ① On meeting end, the server assembles a plain-text transcript and structured Markdown, then auto-POSTs both to the GraphRAG backend for ingestion.

Part 2: The Knowledge Graph

A FastAPI + Neo4j backend ingests transcripts and documents into a persistent, queryable graph — structured by Claude, deduplicated by embedding similarity, and continuously enriched across sessions.

1

Chunk & Extract

Break text into small overlapping chunks, then have Claude pull out the key entities and relationships.

2

Embed & Deduplicate

Convert entities into vectors and merge near-duplicates so the graph stays clean and unified.

3

Debrief Agent

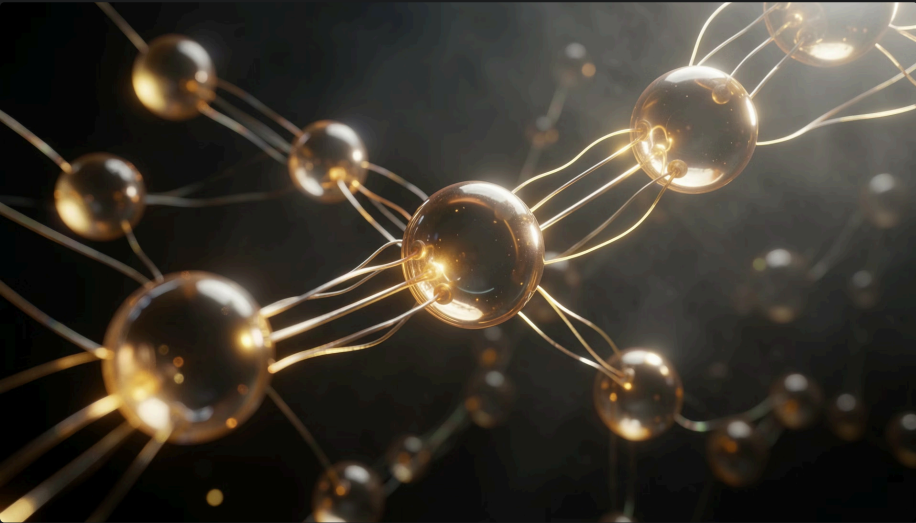
After each meeting, the agent compares new decisions with past context, flags conflicts, and sends a structured summary.

4

Query Modes

Local mode finds nearby facts fast. Global mode looks across communities for broader, higher-level reasoning.

Part 3: The 3D Knowledge Graph Visualization



Rendering Architecture

Three.js + D3 render the graph in real time.

Node Encoding

- CONCEPT and PERSON nodes use warm tones
- ORGANIZATION and TECHNOLOGY nodes use cool tones
- Node size reflects connection degree

Interaction

Click to select, drag to pin, and zoom/orbit to explore the graph.